

Lecture-9
Mathematics 1 (15B11MA111)

CO [C105.1]

Module: Partial Differentiation
Taylor's series for functions of one and two variables

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics”
fifth edition, Narosa publishing house, 2016.

Topics to be covered

- Taylor's Theorem
- Error Estimate
- Exercise

One Variable

It is used to approximate a continuous function in terms of polynomial. It is an extension of Mean-Value Theorem (MVT) which gives a relation between function and the first order derivative.

One Variable

Taylor's Theorem for one variable

Let $f(x)$ be a function which is defined and have continuous derivatives upto $(n + 1)^{th}$ order in some interval I , containing a point a . Then, Taylor's expansion of the function $f(x)$ about the point $x = a$ is given by

$$f(x) = f(a) + \frac{(x - a)}{1!} f'(a) + \frac{(x - a)^2}{2!} f''(a) + \dots \\ + \frac{(x - a)^n}{n!} f^n(a) + R_n(a),$$

where $R_n(a) = \frac{(x - a)^{n+1}}{(n+1)!} f^{n+1}(c)$, is called **the remainder** $a < c < x$ or **error term** of expansion.

Two Variable

Taylor's Theorem for two variable

Let a function $f(x, y)$ be defined in some domain D and have continuous partial derivative upto $(n + 1)^{th}$ order in some neighbourhood of a point $P(x_0, y_0)$ in D . Then, for some point $(x_0 + h, y_0 + k)$ in this neighbourhood, we have Taylor's theorem for two variable

$$\begin{aligned} & f(x_0 + h, y_0 + k) \\ &= f(x_0, y_0) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right) f(x_0, y_0) + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^2 f(x_0, y_0) + \dots \\ & \quad + \frac{1}{n!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^n f(x_0, y_0) + R_n, \end{aligned}$$

$$\text{where } R_n = \frac{1}{(n+1)!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^{n+1} f(x_0 + \theta h, y_0 + \theta k), \quad 0 < \theta < 1.$$

Proof of Taylor's theorem for two variable

I – Way of Proof

Let $x = x_0 + th$ and $y = y_0 + tk$, where t is any parameter in $[0,1]$. Define some $\phi(t)$ as $\phi(t) = f(x, y) = f(x_0 + th, y_0 + tk)$. Using chain rule, we get

$$\begin{aligned}\phi'(t) &= \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} \\ &= \frac{\partial f}{\partial x} \cdot h + k \cdot \frac{\partial f}{\partial y} \\ &= \left(h \cdot \frac{\partial f}{\partial x} + k \cdot \frac{\partial f}{\partial y} \right).\end{aligned}$$

$$\begin{aligned}
\phi''(t) &= \frac{d[\phi'(t)]}{dt} \\
&= \frac{d}{dt} \left(h \cdot \frac{\partial f}{\partial x} + k \cdot \frac{\partial f}{\partial y} \right) \\
&= h \cdot \frac{d}{dt} \left(\frac{\partial f}{\partial x} \right) + k \cdot \frac{d}{dt} \left(\frac{\partial f}{\partial y} \right) \\
&= h \cdot \left(\frac{\partial^2 f}{\partial x^2} \frac{dx}{dt} + \frac{\partial}{\partial y} \frac{\partial f}{\partial y} \frac{dy}{dt} \right) \\
&\quad + k \cdot \left(\frac{\partial^2 f}{\partial y^2} \frac{dy}{dt} + \frac{\partial}{\partial x} \frac{\partial f}{\partial y} \frac{dx}{dt} \right) \\
&= h \cdot \left(\frac{\partial^2 f}{\partial x^2} \cdot h + \frac{\partial^2 f}{\partial x \partial y} \cdot k \right) \\
&\quad + k \cdot \left(k \cdot \frac{\partial^2 f}{\partial y^2} + h \cdot \frac{\partial^2 f}{\partial x \partial y} \right)
\end{aligned}$$

$$\begin{aligned}
\phi''(t) &= h^2 \cdot \frac{\partial^2 f}{\partial x^2} + hk \cdot \frac{\partial^2 f}{\partial x \partial y} + k \cdot \frac{\partial^2 f}{\partial y^2} \\
&\quad + hk \cdot \frac{\partial^2 f}{\partial x \partial y} \\
&= \left(h \cdot \frac{\partial}{\partial x} + k \cdot \frac{\partial}{\partial y} \right)^2 f \\
&\quad \vdots
\end{aligned}$$

Similarly,

$$\phi^{(n+1)}(t) = \left(h \cdot \frac{\partial}{\partial x} + k \cdot \frac{\partial}{\partial y} \right)^{(n+1)} f.$$

Using Taylor's theorem for one variable i.e.,

$$f(x) = f(a) + \frac{(x-a)}{1!}f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$$

Using $\phi(t) = f(x)$

$$\phi(t) = \phi(a) + \frac{(t-a)}{1!}\phi'(a) + \frac{(t-a)^2}{2!}\phi''(a) + \dots$$

Put $t = 1$ and $a = 0$

$$\begin{aligned}\phi(t) &= \phi(0) + \phi'(0) + \frac{1}{2!}\phi''(0) + \dots \\ &\quad + \frac{1}{n!}\phi^n(0) + \frac{1}{(n+1)!}\phi^{(n+1)}(\theta).\end{aligned}\tag{1}$$

$$\phi(0) = f(x_0, y_0)$$

$$\phi(1) = f(x_0 + h, y_0 + k)$$

$$\phi^i(0) = \left(h \cdot \frac{\partial}{\partial x} + k \cdot \frac{\partial}{\partial y} \right)^{(i)} f(x_0, y_0), \quad i = 1, 2, 3, \dots, n$$

$$\phi^{(n+1)}(\theta) = \left(h \cdot \frac{\partial}{\partial x} + k \cdot \frac{\partial}{\partial y} \right)^{(n+1)} f(x_0 + h\theta, y_0 + k\theta), \quad 0 < \theta < 1$$

But above values in (1), we get Taylor's theorem of two variables i.e.,

$$\begin{aligned} f(x_0 + h, y_0 + k) &= f(x_0, y_0) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f(x_0, y_0) \\ &\quad + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f(x_0, y_0) + \dots \end{aligned}$$

Put $x = x_0 + h$, $y = y_0 + k$ in above, we get

$$f(x, y) = f(x_0, y_0) + \left((x - x_0) \frac{\partial}{\partial x} + (y - y_0) \frac{\partial}{\partial y} \right) f(x_0, y_0) \\ + \cdots + R_n,$$

$$\text{where } R_n = \frac{1}{(n+1)!} \left((x - x_0) \frac{\partial}{\partial x} + (y - y_0) \frac{\partial}{\partial y} \right)^{(n+1)} f(\xi, \eta)$$

$$\xi = (1 - \theta)x_0 + \theta x,$$

$$\eta = (1 - \theta)y_0 + \theta y,$$

where $0 < \theta < 1$. When $x_0 = 0$, $y_0 = 0$, we obtain **Maclaurin's Theorem**.

For $n = 1$ and $n = 2$, we get Linear and quadratic polynomial approximations.

II – Way of Proof of Taylor's Theorem of two variables

We have the Taylor's theorem for one variable

$$f(x + h) = f(x) + h \cdot f'(x) + \frac{h^2}{2!} f''(x) + \dots$$

Now let $f(x, y)$ be a function of two independent variables x and y . Keep y constant and expand along x

$$\begin{aligned} f(x + h, y + k) &= f(x, y + k) + h \cdot \frac{\partial}{\partial x} f(x, y + k) \\ &\quad + \frac{h^2}{2!} \frac{\partial^2}{\partial x^2} f(x, y + k) + \dots . \end{aligned} \tag{2}$$

Keeping x constant and expand along y , we get

$$\begin{aligned} f(x, y+k) &= f(x, y) + k \cdot \frac{\partial}{\partial y} f(x, y) \\ &\quad + \frac{k^2}{2!} \frac{\partial^2}{\partial y^2} f(x, y) + \dots . \end{aligned} \tag{3}$$

Put $f(x, y + k)$ from (3) in (2), we get

$$\begin{aligned} f(x + h, y + k) &= \left[f(x, y) + k \cdot \frac{\partial}{\partial y} f(x, y) \right. \\ &\quad \left. + \frac{k^2}{2!} \frac{\partial^2}{\partial y^2} f(x, y) + \dots \right] \\ &\quad + h \cdot \frac{\partial}{\partial x} \left[f(x, y) + k \cdot \frac{\partial}{\partial y} f(x, y) \right. \\ &\quad \left. + \frac{k^2}{2!} \frac{\partial^2}{\partial y^2} f(x, y) + \dots \right] \\ &\quad + \frac{h^2}{2!} \frac{\partial^2}{\partial x^2} \left[f(x, y) + k \cdot \frac{\partial}{\partial y} f(x, y) + \dots \right] \\ &= f(x, y) + \left(h \cdot \frac{\partial}{\partial x} + k \cdot \frac{\partial}{\partial y} \right) f(x, y) + \dots . \end{aligned}$$

Note-I

When $n = 1$, the linear polynomial approximation to $f(x, y)$ is

$$f(x, y) \approx f(x_0, y_0) + (x - x_0)f_x + (y - y_0)f_y,$$

when partial derivatives are evaluated at (x_0, y_0) .

Note-II

When $n = 2$, the quadratic polynomial approximation to $f(x, y)$ is

$$f(x, y) \approx f(x_0, y_0) + (x - x_0)f_x + (y - y_0)f_y \\ + \frac{1}{2!} \{ (x - x_0)^2 f_{xx} + 2(x - x_0)(y - y_0)f_{xy} + (y - y_0)^2 f_{yy} \},$$

when partial derivatives are evaluated at (x_0, y_0) .

Remarks

1. When $(x_0, y_0) = (0, 0)$ then this expansion is called as Maclaurin's expansion.
2. When $\lim_{n \rightarrow \infty} R_n = 0$, we get the Taylor's series expansion of $f(x, y)$ about (x_0, y_0)

Error estimate

The point (ξ, η) or the value of θ given in the expansion is not known, therefore, error term can't be evaluated exactly.

We try to find a bound of the error in a rectangular region $R : |x - x_0| < \delta_1, |y - y_0| < \delta_2$, assuming that all the partial derivatives are continuous in this region.

When, $n = 1$, the error term is

$$R_1 = \frac{1}{2!} \{ (x - x_0)^2 f_{xx} + 2(x - x_0)(y - y_0) f_{xy} + (y - y_0)^2 f_{yy} \}$$

where partial derivatives are evaluated as

$(\xi, \eta) = [(1 - \theta)x_0 + \theta x, (1 - \theta)y_0 + \theta y], 0 < \theta < 1$, hence we find

$$|R_1| \leq \frac{1}{2!} \{ |x - x_0|^2 |f_{xx}| + 2|x - x_0| |y - y_0| |f_{xy}| + |y - y_0|^2 |f_{yy}| \}$$

If we assume that $B = \max\{|f_{xx}|, |f_{xy}|, |f_{yy}|\}$ for all (x, y) in R .

Then

$$\begin{aligned}|R_1| &\leq \frac{B}{2} \left\{ |x - x_0|^2 + 2|x - x_0||y - y_0| + |y - y_0|^2 \right\} \\ &= \frac{B}{2} \left\{ |x - x_0| + |y - y_0| \right\}^2 \\ &\leq \frac{B}{2} (\delta_1 + \delta_2)^2.\end{aligned}$$

Hence $|R_1|$ is called the max absolute error in the linear approximation of $f(x, y)$ about (x_0, y_0) .

Similarly for $n = 2$, the error term is

$$\begin{aligned} R_2 &\leq \frac{1}{3!} \{ (x - x_0)^3 f_{xxx} + 3(x - x_0)^2(y - y_0)f_{xxy} \\ &\quad + 3(x - x_0)(y - y_0)^2 f_{xyy} + (y - y_0)^3 f_{yyy} \} \end{aligned}$$

where partial derivatives are evaluated as

$(\xi, \eta) = [(1 - \theta)x_0 + \theta x, (1 - \theta)y_0 + \theta y], 0 < \theta < 1$, hence we find

$$\begin{aligned} |R_2| &\leq \frac{B}{6} \{ |x - x_0|^3 + 3 |x - x_0|^2 |y - y_0| + 3 |x - x_0| |y - y_0|^2 \\ &\quad + |y - y_0|^3 \} \end{aligned}$$

If we assume that $B = \max\{|f_{xxx}|, |f_{xxy}|, |f_{xyy}|, |f_{yyy}|\}$. The same concept can be further extended for three variables.

Example

Find the linear and quadratic Taylor's series polynomial approximations to the function $f(x, y) = 2x^3 + 3y^3 - 4x^2y$ about $(1, 2)$. Obtain the max absolute error in the region $|x - 1| < 0.01$ and $|y - 2| < 0.01$.

Solution:

Here $f(x, y) = 2x^3 + 3y^3 - 4x^2y$ and $f(1, 2) = 18$

$$\begin{aligned} f_x &= 6x^2 - 8xy, & f_{xx}(1, 2) &= -4. \\ f_x(1, 2) &= -10. & f_{yy} &= 18y, \\ f_y &= 9y^2 - 4x^2, & f_{yy}(1, 2) &= 36. \\ f_y(1, 2) &= 32. & f_{xy} &= -8x, \\ f_{xx} &= 12x - 8y, & f_{xy}(1, 2) &= -8. \end{aligned}$$

$$f_{xxx} = 12,$$

$$f_{xxy} = -8,$$

$$f_{xyy} = 0,$$

$$f_{yyy} = 18.$$

Linear approximation is,

$$\begin{aligned} f(x, y) &= f(1, 2) + (x - 1)f_x(1, 2) + (y - 2)f_y(1, 2) \\ &= 18 - 10(x - 1) + 32(y - 2). \end{aligned}$$

max absolute error is

$$\begin{aligned} |R_1| &\leq \frac{B}{2} \left\{ |x - 1| + |y - 2| \right\}^2 \\ &\leq \frac{B}{2} [0.01 + 0.1]^2 \\ &= 0.00605B. \end{aligned}$$

and $B = \max\{|f_{xx}|, |f_{xy}|, |f_{yy}|\}$

$$\begin{aligned}\max\{|f_{xx}|\} &= \max\{|12x - 8y|\} \\ &= \max\{|12(x - 1) - 8(y - 2) + 4|\} \\ &\leq \max\left[12|x - 1| + 8|y - 2| + 4\right] \\ &= 4.92.\end{aligned}$$

$$\begin{aligned}\max\{|f_{yy}|\} &= \max\{|18y|\} \\ &= \max\{|18(y - 2) + 36|\} \\ &\leq \max\left[18|y - 2| + 36\right] \\ &= 1.8 + 36 = 37.8.\end{aligned}$$

$$\begin{aligned}
\max\{|f_{xy}|\} &= \max\{|-8x|\} \\
&= \max\{|8(x-1)+8|\} \\
&\leq \max[8|x-1|+8] \\
&= 0.8+8=8.08.
\end{aligned}$$

Hence $B = 37.8$ and $|R_1| \leq 0.00605(37.8) \simeq 0.23$.

The quadratic polynomial approximation is

$$\begin{aligned}
f(x, y) &= 18 - 10(x-1) + 32(y-2) \\
&\quad - 2[(x-1)^2 + 4(x-1)(y-2) - 9(y-2)^2].
\end{aligned}$$

and max absolute error is

$$\begin{aligned}|R_2| &\leq \frac{B}{6} \left\{ |x - 1| + |y - 2| \right\}^3 \\ &\leq \frac{B}{6} (0.11)^3.\end{aligned}$$

and

$$\begin{aligned}B &= \max\{|f_{xxx}|, |f_{xxy}|, |f_{xyy}|, |f_{yyy}|\} \\ &= \max\{12, 0, 8, 18\} \\ &= 18.\end{aligned}$$

$$|R_2| \leq \frac{18}{6} (0.00133) = 0.004.$$

Exercise

Expand $f(x, y) = 21 + x - 20y + 4x^2 + xy + 6y^2$ in Taylor's series of max order about $(-1, 2)$.

Ans:

$$f(x, y) = 6 - 5(x + 1) + 3(y - 2) + 4(x + 1)^2 + (x + 1)(y - 2) + 6(y - 2)^2$$

Thank You